

Rahul Rawat

Data Science Intern

PERSONAL DETAILS

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Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on experience in Python and PySpark, dashboard driven analytics, and cross functional data delivery on real production style data. I have shipped a real time flight tracking pipeline processing over **128 thousand** records with PySpark on Google Cloud, an automated Bronze to Silver to Gold BigQuery medallion architecture with a leakage free BigQuery ML classifier and a five page Looker Studio dashboard, and an interactive Tableau dashboard with dynamic Set Actions and parameter driven analytics for non technical stakeholders. Comfortable in Python, PySpark, SQL, dashboard tooling that transfers directly to Power BI, and clear stakeholder communication, I am the right fit for the six month Data Science Praktikum at Johnson and Johnson in Norderstedt.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026 Heidelberg	eRay GmbH Data Science Researcher - Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg. - Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. - Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors. - Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction. - Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.
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Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

PERSONAL PROJECTS

2025

Real Time Flight Tracking Data Pipeline

- Collected live flight positions from the OpenSky Network API every **30 seconds** and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than **128 thousand** records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every **15 minutes**, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops **4.4 times** in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

Technologies: *Python, PySpark, BigQuery, dbt, Apache Airflow, GCP Dataproc and GCS, Tableau with TabPy*

2025

Movie Analytics and ML Pipeline on GCP

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing.

Technologies: *GCP BigQuery, Cloud Run, GCS, Cloud Scheduler, BigQuery ML, Python, SQL, Looker Studio*

2025

Fast Food Nutritional Analyzer and Meal Simulator

- Built a dynamic shopping cart inside Tableau, measured by the fact that end users can select scatter plot points and instantly total calories, fat, and protein for a simulated meal, using Tableau Set Actions as the interaction layer.
- Implemented parameter driven analytics with a dynamic Y axis tied to a user controlled goal parameter for muscle gain versus weight loss, measured by consistent axis switching without dashboard reloads, using a CASE statement over the parameter.
- Authored complex order of operation IF and THEN calculated fields for logical grouping and custom flags such as an Is It A Trap flag for deceptive high fat and high calorie items, measured by manual spot checks against the source nutrition data.
- Designed a two tier layout combining an executive macro view and a granular food finder using a colour blind safe dark mode palette, measured against a target of reduced time to insight for a non technical stakeholder.
- Link to Tableau: <https://sl1nk.com/n7ojnln>

Technologies: *Tableau with Set Actions, parameters and calculated fields, data storytelling and UI or UX*

EDUCATION

Apr 2025 to Present Heidelberg	M.Sc. Data Science and Analytics SRH University of Applied Sciences Heidelberg, GPA 1.9
2019 to 2023 Greater Noida, India	Bachelor of Technology in Computer Science GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023 Greater Noida, India	Bachelor Thesis, Diabetes Prediction Using Machine Learning GL Bajaj Institute of Technology and Management, IEEE style paper - Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable. - Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit. - Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class. - Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance. Technologies: <i>Python, scikit learn, Pandas, Seaborn</i>
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CERTIFICATIONS

- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native

Mannheim, 20 July 2026
Rahul Rawat